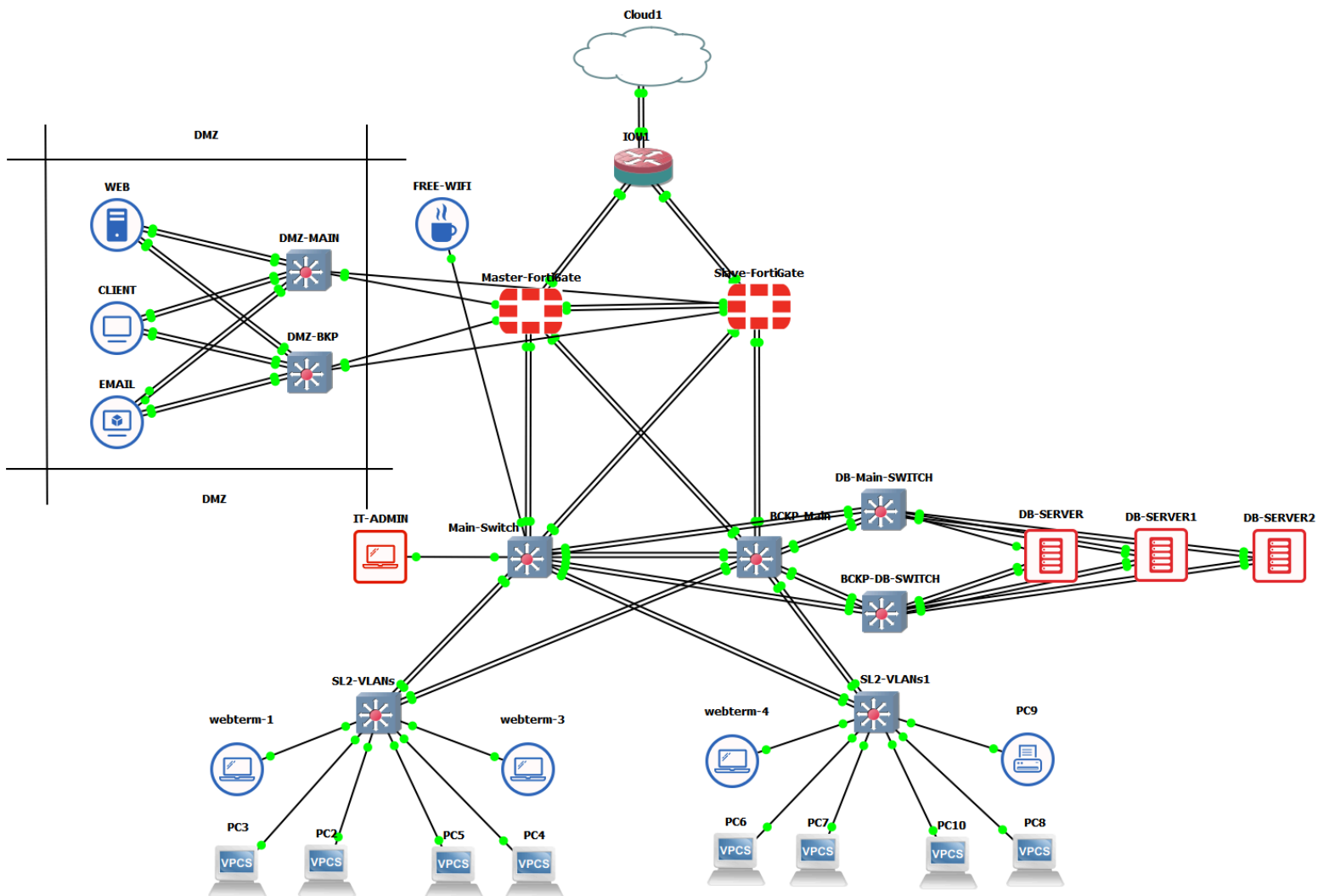
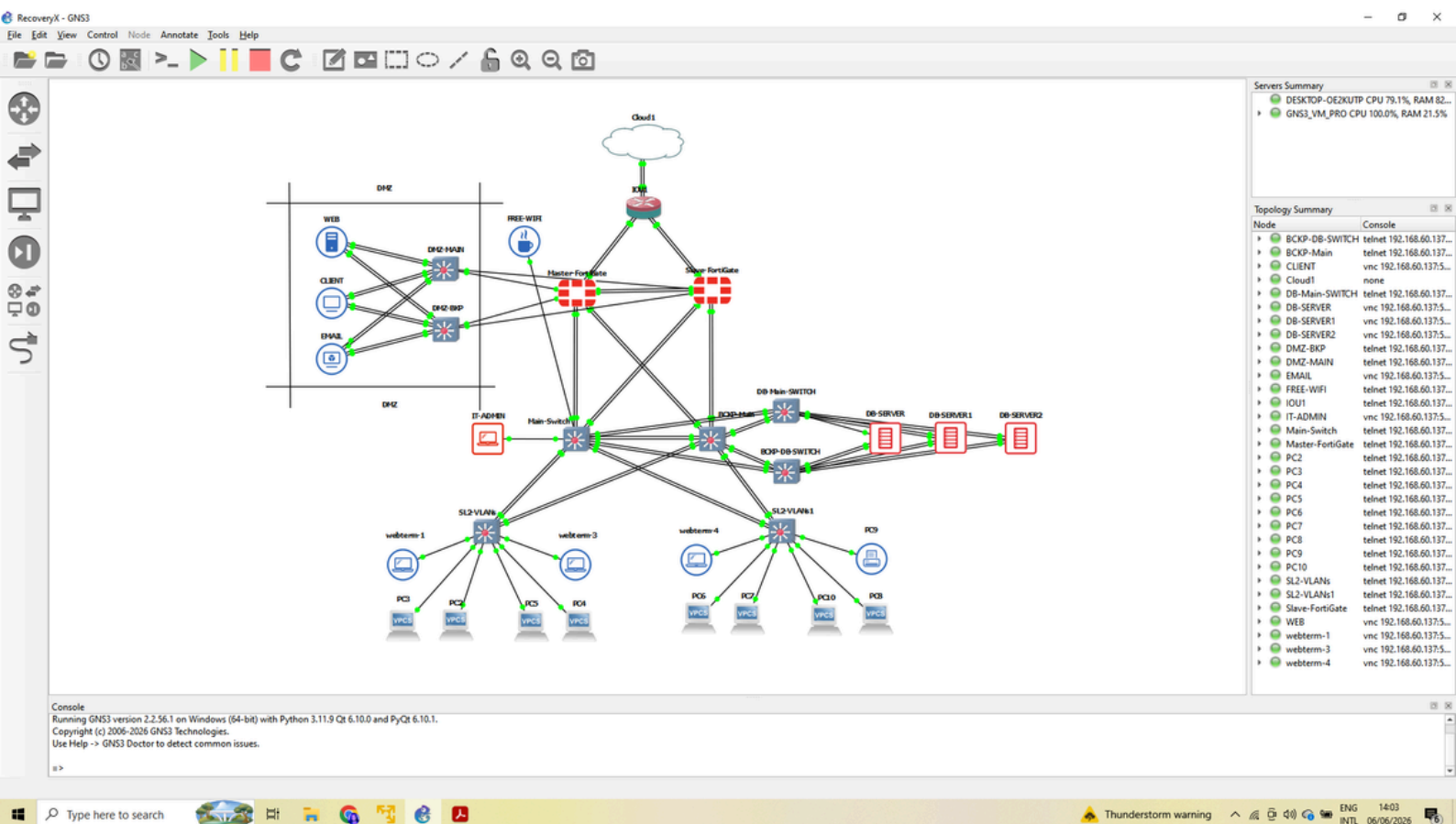


WELCOME TO MY PROJECT HIGH-AVAILABILITY ENTERPRISE INFRASTRUCTURE

THE VISION & JOURNEY

This is a production-ready, highly redundant enterprise topology built completely from scratch inside the GNS3 laboratory. Started on March 2, 2026 and finalised on 6 June, this project represents an intensive 3-month hands-on engineering designed to eliminate any Single Point of Failure (SPOF) while enforcing rigorous zero-trust security.





PRODUCTION BLUEPRINT & TOPOLOGY

Perimeter & Security Tier

- 2x FortiGate Firewalls (HA Cluster): Configured in Active-Passive High Availability mode to guarantee uninterrupted gateway traffic and seamless failover.
- Segmented Zones: Complete isolation between Internal LANs, DMZ services, and Public zones.

Core & Network Nucleus

- 2x Multilayer Layer 3 Switches: * Main-Switch acts as the true heart and computing nucleus of the entire infrastructure.
 - BCKP-Main acts as the hot-standby unit, ready to take over line-rate routing instantly.
- 6x Layer 2 Switches: Deployed strictly as a high-speed transport layer and stack cables, moving multi-VLAN traffic across the campus matrix.

The DMZ (Demilitarised Zone)

- 3x Dedicated Tier Servers: Separated into Web, Client, and Email services.
- Redundant Plumbing: Every DMZ server is multi-homed, dual-connected simultaneously to a primary L2 Switch and a secondary L2 Backup switch to survive physical link failures.

The Quorum Data Centre (The Brain)

- 3-Node Backend Server Cluster (Quorum Design): Engineered explicitly with 3 production nodes to eliminate the catastrophic *Split-Brain* scenario in clustering.

- High-Throughput Provisioning: Every cluster node uses dual-bonded network links directly connected to the network core.

Special Purpose Zones

- FREE-WIFI: Completely isolated guest zone with direct-to-internet routing only.
- IT-ADMIN: A dedicated, hyper-secured management zone allowing out-of-band access directly to the source of any infrastructure incident.

THE LOGICAL SUBNETTING & VLAN MATRIX

(Extracted directly from Core FortiGate Interface Engine)

VLAN ID	Interface Name	IP / Netmask	Security Profile & Role
VLAN 10	Lan10_IT	192.168.10.1/24	Internal Core IT Operations Subnet
VLAN 20	HR_DEP	192.168.20.1/24	Human Resources - Isolated Traffic
VLAN 30	SALES_DEP	192.168.30.1/24	Sales Department - Rate-limited Internet Access
VLAN 40	Other_DEP	192.168.40.1/24	General Corporate Internal Traffic
VLAN 66	FREE_WIFI	192.168.66.1/24	Guest Hotspot - NAT Enabled / Zero LAN Access
VLAN 99	IT_ADMIN	192.168.99.1/24	Secure Management / Direct Source Troubleshooting
VLAN 100	DB_SERVERS	192.168.100.1/24	Production Data Cluster - Protected Database Zone

FortiGate VM64-KVM

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New

Edit

Delete

Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
802.3ad Aggregate 10							
AGG-CORE	802.3ad Aggregate	port4 port3	0.0.0.0/0.0.0.0	PING HTTPS			8
DB-SERVERS	VLAN		192.168.100.1/255.255.255.0	PING HTTPS			3
FREE_WIFI	VLAN		192.168.66.1/255.255.255.0	PING HTTPS			2
HR-DEP	VLAN		192.168.20.1/255.255.255.0	PING HTTPS	2	192.168.20.2-192.168.20.254	2
IT_ADMIN	VLAN		192.168.99.1/255.255.255.0	PING HTTPS			2
Lan10-IT	VLAN		192.168.10.1/255.255.255.0	PING HTTPS	2	192.168.10.2-192.168.10.254	2
Other_DEP	VLAN		192.168.40.1/255.255.255.0	PING HTTPS		192.168.40.2-192.168.40.254	2
SALES_DEP	VLAN		192.168.30.1/255.255.255.0	PING HTTPS		192.168.30.2-192.168.30.254	2
AGG-CORE-BKP	802.3ad Aggregate	port5 port6	0.0.0.0/0.0.0.0	PING HTTPS			1
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254	2

CORE PROTOCOLS & HARDENING METHODOLOGIES

"I don't build flat networks. I engineer secure fortresses."

- **LACP Dynamic EtherChannel (802.3ad):** Configured actively across FortiGate firewalls and core switches to automate link bundling, maximising throughput and enabling automatic state recovery if a cable is disconnected.
- **802.1Q Tagged Trunking:** Enforced on all inter-switch trunk links to transport multi-tenant enterprise subnets with zero cross-VLAN leakage.
- **PVST+ (Per-VLAN Spanning Tree):** Eliminates Layer 2 switching loops across the redundant topology while maintaining the Main-Switch as the strategically elected Root Bridge.
- **Layer 4 Microsegmentation:** The public Web Server in the DMZ can query the production database cluster **strictly via TCP Port 3306**. Any lateral movement or unauthorised port probes are immediately blocked and logged.
- **Stealth Topology Design (ICMP Silent Target):** The database server environment is configured to drop all ICMP echo requests silently. It acts as a black hole to network sweeps, preventing unauthorised asset discovery. The only node permitted to ping the database is the verified IT_ADMIN IP block.
- **Banner Grabbing Mitigation:** Service application headers, OS release signatures, and default software banners have been explicitly masked. This denies attackers the critical intelligence needed during the *Reconnaissance* phase to launch targeted version exploits.

LOGICAL INTERFACE & LINK AGGREGATION MATRIX (FORTIGATE)

Interface Name	Interface Type	Member Physical Ports	IP / Subnet Mask	Role in the Infrastructure
AGG-CORE	802.3ad Aggregate	port3, port4	0.0.0.0/0.0.0.0	Primary aggregated link (LACP) to the Core network
BKP-AGG-CORE	802.3ad Aggregate	port5, port6	0.0.0.0/0.0.0.0	Backup aggregated link to the secondary Core switch
DMZ_WEB_SERVER	Physical Interface	port9	192.168.77.1/24	Dedicated segment for public Web, Client, Email servers
The_Way_Out_WAN	Physical Interface	port1	192.168.60.200/24	Direct connection to the external Internet (Cloud)

FortiGate VM64-KVM

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New

Edit

Delete

Integrate Interface

Search

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP
802.3ad Aggregate 10						
Physical Interface 6						
DMZ_WEB_SERVER (port9)	Physical Interface		192.168.77.1/255.255.255.0	PING HTTPS		
port2	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			
port8	Physical Interface		0.0.0.0/0.0.0.0			
port10	Physical Interface		0.0.0.0/0.0.0.0			
The_Way_Out_WAN (port1)	Physical Interface		192.168.60.200/255.255.255.0	PING HTTPS SSH		

FORTIGATE FIREWALL POLICY MATRIX

- **Baseline Security Posture:** Implicit Deny (**DROP ALL**). Any network traffic not explicitly defined below is immediately dropped at the kernel level.

Policy Name	Source Interface	Destination Interface	Allowed Services	Action / NAT
DMZ_to_DB	DMZ_WEB_SERVER	AGG-CORE (VLAN 100)	MYSQL (Port 3306)	ACCEPT / NAT Disabled
FREE-WIFI	AGG-CORE (VLAN 66)	The_Way_Out_WAN	ALL	ACCEPT / NAT Enabled
Admin_to_DM	AGG-CORE (VLAN 99)	AGG-CORE (VLAN 100)	ALL_ICMP, PING	ACCEPT / NAT Disabled

Q

HA: Primary

+ Create New

Edit

Delete

Q Policy Lookup

Search

Q

Export

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log
DMZ_WEB_SERVER (port9) → DB-SERVERS ⓘ								
DMZ_to_DB	all	DB-SERVERS address	always	MYSQL	✓ ACCEPT	⛔ Disabled	SSL no-inspection	UTM
FREE_WIFI → The_Way_Out_WAN (port1) ⓘ								
FREE-WIFI_to_Internet	all	all	always	ALL	✓ ACCEPT	✓ Enabled	SSL no-inspection	UTM
IT_ADMIN → DB-SERVERS ⓘ								
Admin_to_DM	IT_ADMIN address	DB-SERVERS address	always	ALL_ICMP PING	✓ ACCEPT	⛔ Disabled	SSL no-inspection	UTM
Implicit ⓘ								
Implicit Deny	all	all	always	ALL	⛔ DENY			⛔ Disabled

TROUBLESHOOTING LOGS (INCIDENT MANAGEMENT)

###CASE 1: Cisco IOU Kernel Crash (Segfault -11)

- * **Symptom:** Both Core switches entered a continuous crash loop inside GNS3, completely freezing the CLI consoles.
- * **Root Cause:** A native memory allocation bug within Cisco IOU images triggered during a hot transition of a Port-channel from Layer 3 (Routed) to Layer 2 (Switchport).

*** Resolution:** Performed an offline recovery by modifying the `startup-config.cfg` file via GNS3's `Edit config` tool. The corrupted interface configuration block was completely wiped from NVRAM. Executed a clean cold boot and pre-injected the LACP configurations directly into the boot files while physical interfaces were kept in a `shutdown` state, successfully bypassing the faulty dynamic negotiation phase.

```
interface Ethernet1/2
switchport trunk encapsulation dot1q
switchport mode trunk
duplex auto
channel-group 2 mode active
!
interface Ethernet1/3
switchport trunk encapsulation dot1q
switchport mode trunk
duplex auto
channel-group 2 mode active
!
```

###CASE 2: EtherChannel Port Mismatch (Po10 Stuck in RD Mode)

*** Symptom:** The virtual bundle failed to carry multi-VLAN traffic, and physical ports refused to bundle correctly into the operational group.

*** Root Cause:** The switch operating system initialised the logical interface as a Layer 3 Router Interface (Flag `R`) and kept it in a down state (Flag `D`), blocking outbound LACP negotiation packets.

*** Resolution:** Forced the physical and logical interfaces into pure Layer 2 mode using the `switchport` command, cleared the stale channel-group configurations, and issued the `no shutdown` command simultaneously on both connecting nodes to force the EtherChannel into a stable `SU` (In Use) state.

```
Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
2      Po2(SU)          LACP       Et1/2(P)  Et1/3(P)
3      Po3(SU)          LACP       Et2/2(P)  Et2/3(P)
10     Po10(SU)         -          Et0/0(P)  Et0/1(P)

Main-Switch#
```

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###CASE 3: DHCP IP Allocation Failure on Access Workstations

*** Symptom:** Virtual PCs connected to VLAN 10 and VLAN 20 failed to acquire automatic IP addresses from the FortiGate DHCP pools when executing the `ip dhcp` command.

* **Root Cause:** Missing explicit Trunk encapsulation permissions on the Port-channel link between Core and Access layers. Because specific VLAN IDs were not allowed, DHCP broadcast discovery packets were dropped at the distribution plane.

* **Resolution:** Applied the `switchport trunk allowed vlan all` command across the logical `Port-channel` interfaces on both ends. Configured correct Access port bindings on the edge switch to guarantee flawless 802.1Q tag propagation directly to the FortiGate subinterfaces.

```
PC2> show ip

NAME       : PC2[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20150
RHOST:PORT : 127.0.0.1:20151
MTU        : 1500

PC2> ip dhcp
DDORA IP 192.168.10.3/24 GW 192.168.10.1

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.168.10.3/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8 1.1.1.1
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 604796, 604800/302400/529200
MAC        : 00:50:79:66:68:00
LPORT      : 20150
RHOST:PORT : 127.0.0.1:20151
MTU        : 1500

PC2> 
```

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